

What Can I Do to Achieve SDG Goal 4?

Quality Education through AI Career Guidance

Eita MATSUURA

Nagaoka University of Technology

Gigaku Innovation and Creativity

24336684



About Me

My Background and Research

- Master's student at **Nagaoka University of Technology**
- Research: **An AI system to support high school students in choosing their careers**
 - Technology: SGA (Simulation with Generative Agents)
 - Students fill out a form → AI suggests personalized career paths

How the system works

- Reads each student's interests and values
- Suggests career paths from age 18 to 30
- Students choose the ideas they like 

Results so far

- **245 students** at one high school participated
- About **1 in 3** AI suggestions were accepted
- Students showed wider career awareness 

SDG Goal 4: Quality Education

What does Goal 4 aim for?

Sub-targets related to my research:

Sub-target	Content
4.4	Give people the skills needed for good jobs
4.5	Remove education inequality between regions and groups
4.7	Help students understand their role in society and future

Goal 4 is not just about school — it aims to help every person find their own path in life

center

"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all"

The Problem I Want to Solve

Career Education in Japan is Not Equal ⚠

Students in rural areas have less support

- Small schools have fewer teachers and resources
- Career counselors are often not available
- Students from rural areas tend to have lower university enrollment rates

Teachers don't have enough time

- One teacher handles 30–40 students
- Individual career advice is almost impossible
- Many students graduate without a clear picture of their future

Where you grow up should not decide the quality of your career education

How My Research Connects to Goal 4

My AI System Directly Solves This Problem

Current situation ❌

- Quality of career guidance depends on the teacher
- Schools in big cities have more resources
- Students in rural areas get less support
- Individual support for 40 students is not realistic

With my AI system ✅

- Any student, anywhere can get personalized career suggestions
- Works with just a computer and internet connection
- No extra burden on teachers
- Every student gets different, personalized suggestions

My research is a small step toward Goal 4 — but I want to go even further 🎯

What I Can Do ①

Expand the System to More Schools

Short-term goals:

- Test the system at more schools, especially **rural schools** 🏫
- Provide it **for free** to schools with small budgets
- Improve suggestion accuracy by collecting teacher feedback

Connection to Goal 4:

- Right now, only one school in Nagaoka can use it
- By expanding, thousands of students can get equal support
- Reduces the education gap between urban and rural areas

My dream: By 2030, every high school student in Japan can use this system

What I Can Do ②

My Bigger Dreams for the Future 💡

Make it multilingual

- Support English, Chinese, Vietnamese, and more 🌐
- International students living in Japan can also use it
- Directly connected to Goal 4's idea of "education for ALL"

Build a teacher support dashboard

- Visualize which students need more help
- AI and teachers work together — AI does not replace teachers 🤝

Share the research openly

- Publish findings so other researchers can build on it
- Contribute to global education AI research

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s243366@stn.nagaokaut.ac.jp

24336684

